DD2424 Assignment 1

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1 Checking analytical gradient

In this assingment I had to check my analytical gradient against PyTorch's ground truth gradient that is computed with the given function `ComputeGradsWithTorch`. To do this I checked that the gradients hade the same size, and then measured that the maximum absolute difference and relative error both were $< 1e-6$. The obtained results were:

```
Comparing W:
shape my_grads   : (10, 10)
shape torch_grads: (10, 10)
max absolute difference: 1.1102230246251565e-16
max relative error     : 3.1506227451020938e-15

Comparing b:
shape my_grads   : (10, 1)
shape torch_grads: (10, 1)
max absolute difference: 5.551115123125783e-17
max relative error     : 1.190200411005617e-16
```

Thus the code for the analytical gradient was made sure to be correct.

2 Training with different parameters

2.1 `lam=0, n_epochs=40, n_batch=100, eta=.1`

These parameter settings gave the loss and cost function seen in Figure 1 and achieved 27.85% accuracy on the test set.

2.2 `lam=0, n_epochs=40, n_batch=100, eta=.001`

These parameter settings gave the loss and cost function seen in Figure 3 and achieved 39.13% accuracy on the test set.

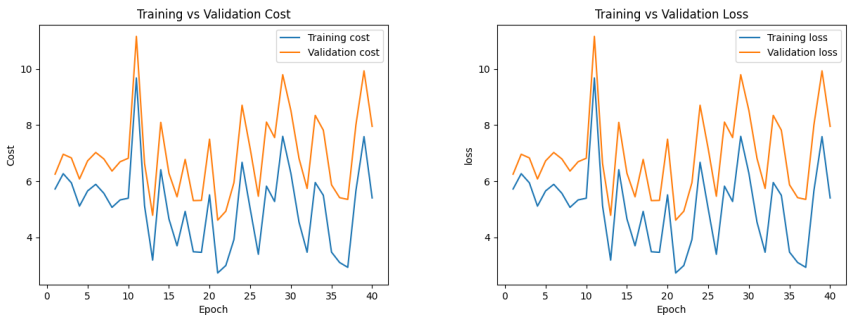

Figure 1: Training and validation cost (left) and loss (right) over epochs for parameters: `lam=0, n_epochs=40, n_batch=100, eta=.1`

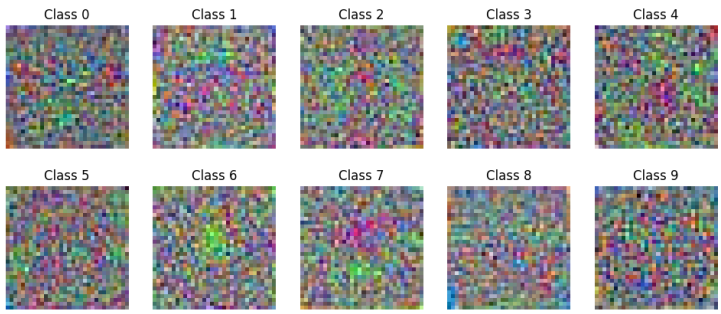

Figure 2: learnt weight matrix for parameters: for parameters: `lam=0, n_epochs=40, n_batch=100, eta=.1`

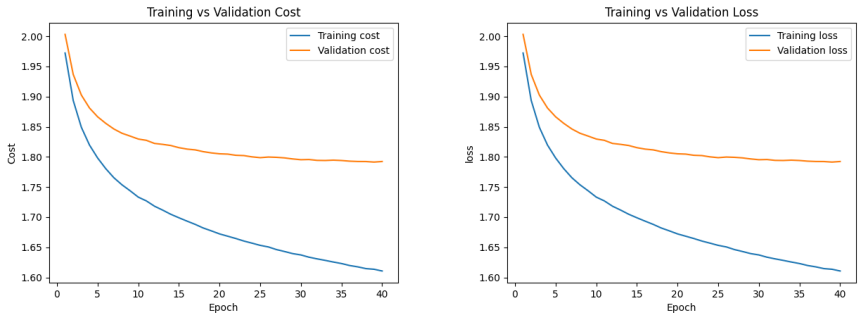

Figure 3: Training and validation cost (left) and loss (right) over epochs for parameters: `lam=0, n_epochs=40, n_batch=100, eta=.001`

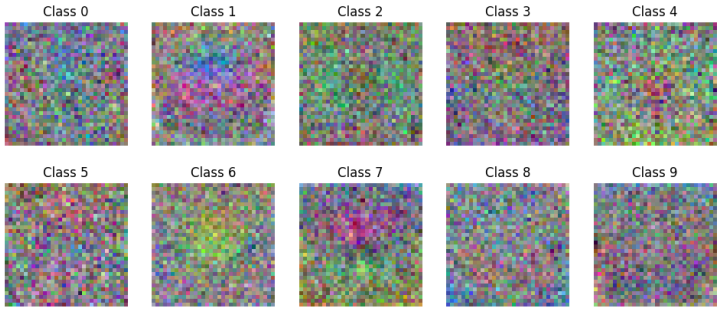

Figure 4: learnt weight matrix for parameters: for parameters: `lam=0, n_epochs=40, n_batch=100, eta=.001`

2.3 `lam=0.1, n_epochs=40, n_batch=100, eta=.001`

These parameter settings gave the loss and cost function seen in Figure 5 and achieved 39.22% accuracy on the test set.

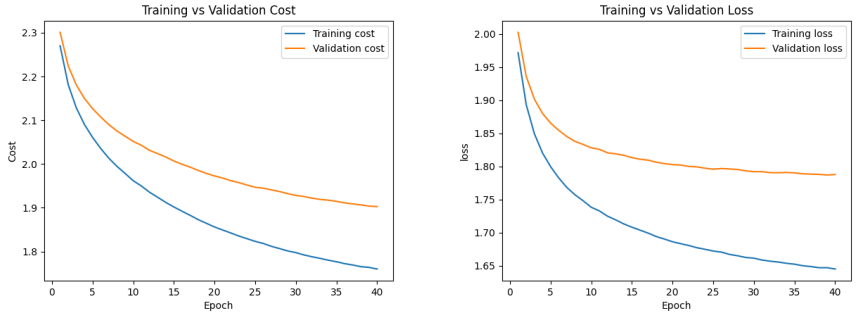

Figure 5: Training and validation cost (left) and loss (right) over epochs for parameters: `lam=0.1, n_epochs=40, n_batch=100, eta=.001`

2.4 `lam=1, n_epochs=40, n_batch=100, eta=.001`

These parameter settings gave the loss and cost function seen in Figure 7 and achieved 37.22% accuracy on the test set.

2.5 Comments

Increasing the regularization parameter reduces overfitting by penalizing larger weights, which often results in a more general model. But excessive regulariza-

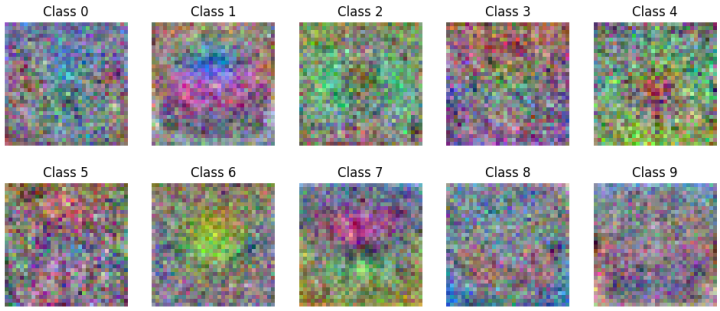

Figure 6: learnt weight matrix for parameters: for parameters: `lam=0.1, n_epochs=40, n_batch=100, eta=.001`

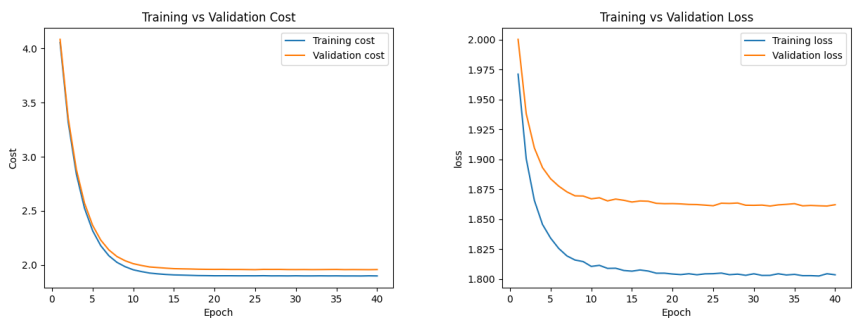

Figure 7: Training and validation cost (left) and loss (right) over epochs for parameters: `lam=1, n_epochs=40, n_batch=100, eta=.001`

tion constrains the model too much which can lead to underfitting leading to poor performance.

The learning rate controls how large parameter updates are during training. Too large learning rate causes unstable training or a model that never converges. On the other hand, if the value is too small the convergence will be very slow which results in suboptimal results in a fixed amount of epochs.

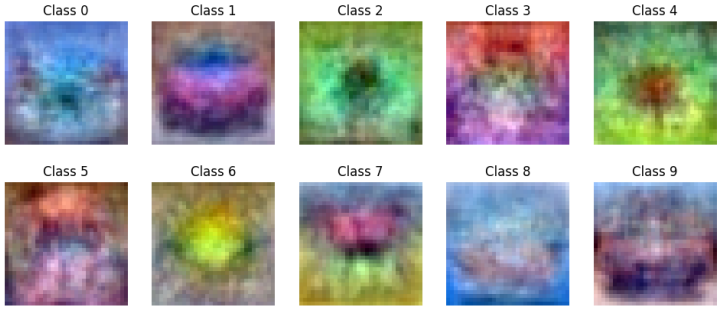

Figure 8: learnt weight matrix for parameters: for parameters: `lam=1, n_epochs=40, n_batch=100, eta=.001`